MTH1003

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

May 2024

Mathematical Modelling

Module Leader: Andrew Gilbert and Jemma Shipton

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) A particle of mass $m = 1$ moves under the action of the force given by

$$\boldsymbol{F}(t) = t\,\boldsymbol{i} + \cos(t)\,\boldsymbol{j} + e^t\,\boldsymbol{k}.$$

At time $t = 0$ the particle's position is $\boldsymbol{r}(0) = 2\boldsymbol{j}$ and its velocity is $\boldsymbol{v}(0) = \boldsymbol{0}$. Find the velocity $\boldsymbol{v}(t)$ and position $\boldsymbol{r}(t)$ for general times t. (8)

(b) Consider the following short Matlab code

```
n = 4 ;
m = 0 ;
for j = 1:n
   m = m + j^2 * sin(j*pi/4)^2 ;
end
disp(m)
```

What value will be displayed when the code is executed? (6)

(c) A ball is projected from the origin with speed V at an angle α to the horizontal. Taking the acceleration due to gravity to be g and neglecting air resistance, give the ball's horizontal and vertical coordinates x and z as functions of time t. Eliminate time to obtain the equation for the trajectory.

Hence find the location $x = R$ where the ball lands, as a function of V, α and g. What angle α maximises the range R and what is this maximum value? (12)

(d) A particle with position $x(t)$ undergoes damped harmonic motion governed by

$$\ddot{x} + 2k\dot{x} + \omega^2 x = 0,$$

with initial conditions $x = 2$, $\dot{x} = 0$ at $t = 0$. For the two cases

$$\begin{aligned} &\text{(I)} \quad k = 1, \quad \omega^2 = 5, \\ &\text{(II)} \quad k = 3, \quad \omega^2 = 5, \end{aligned}$$

obtain the solution $x(t)$ and state what kind of damping occurs. (12)

(e) For the dynamical system

$$\dot{x} = (x-1)(x^2+y^2-1), \quad \dot{y} = x^2 - y^2,$$

sketch the nullclines, and hence or otherwise find all of the equilibrium points.

Classify the type and stability of the fixed point at $x = 0$, $y = 0$ for the dynamical systems

$$\text{(I)} \quad \dot{x} = x + y, \quad \dot{y} = x - y,$$
$$\text{(II)} \quad \dot{x} = x - 2y, \quad \dot{y} = 2x + y.$$

(12)

[**50**]

SECTION B

2. An elastic string with modulus of elasticity λ and length $4l_0$ is taken and three identical particles of mass m are attached to it, one quarter, one half, and three quarters of the way along it. The ends of the string are then attached to fixed points A and B a distance $8l_0$ apart on a smooth horizontal table.

   (a) Draw the configuration described above when the particles are in a state of equilibrium. (2)

   (b) Define the displacements of the three particles from their equilibrium locations as x, y and z, positive in the direction from A to B. Sketch this general configuration, including tension forces. Hence derive the equation of motion for each particle.

   Show that the equations governing this coupled system may be written in matrix form as

$$\begin{pmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{pmatrix} = k^2 \begin{pmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix},$$

   where $k = \sqrt{\lambda/ml_0}$. (10)

   (c) Show that seeking a solution proportional to e^{pt} of the form

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = e^{pt} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

   (with α, β, γ constants) implies that $\mu = p^2/k^2$ is an eigenvalue of the coefficient matrix above. Show that one normal mode has

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix},$$

   and find the corresponding frequency q_1 with $p = \pm iq_1$. (5)

   (d) Show that there is a second normal mode in which the middle particle is at rest, while the first and third particles move in opposite directions with equal amplitude, and find its frequency q_2 with $p = \pm iq_2$. (5)

   (e) Using the fact that the trace of a matrix is the sum of its eigenvalues, find the frequency q_3 for the third normal mode of this system. (3)

   [**25**]

3. (a) A particle of mass $m = 1$ moves along the x-axis under a conservative force given by the potential

$$V(x) = 3x^2(x^2 - 2).$$

(i) Find $V'(x)$ and hence the force $F(x)$ associated with this potential. Also find $V''(x)$ and hence obtain the positions of equilibrium points, their stability, and the frequencies of small oscillations about any stable equilibria.
Sketch the potential $V(x)$ as a function of x.

(ii) Suppose a particle is projected from the right-most stable equilibrium with velocity $\dot{x} = U$. Write down the energy equation and find the total energy E of the particle. Determine the different types of behaviour that can occur, giving the ranges of the energy E and the initial U for each. (15)

(b) (i) Give the Taylor series approximations for $f(x + h)$ and for $f(x-h)$ in terms of f and its derivatives evaluated at the point x. Hence obtain a centred finite difference approximation for $f''(x)$ in terms of $f(x)$, $f(x - h)$ and $f(x + h)$. What is the order of the error in your approximation?

(ii) Now give the Taylor series approximation for $f(x + 3h)$ in terms of f and its derivatives evaluated at x. Obtain a finite difference approximation for $f''(x)$ in terms of $f(x)$, $f(x + h)$ and $f(x + 3h)$. What is the order of the error in your approximation? (10)

[**25**]

4. (a) You are **given** the equation of motion in polar coordinates (r, θ) for a particle of mass m moving under a central force $\boldsymbol{F}(r) = mf(r)\hat{\boldsymbol{r}}$ as:

$$(\ddot{r} - r\dot{\theta}^2)\hat{\boldsymbol{r}} + \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)\hat{\boldsymbol{\theta}} = f(r)\hat{\boldsymbol{r}},$$

where $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\theta}}$ are unit vectors in the radial and azimuthal directions respectively.

(i) Define the angular momentum h per unit mass and show that it is conserved.

(ii) Eliminate $\dot{\theta}$ in terms of h and r, and hence re-express the radial component of the equation of motion in terms of $u = 1/r$ with independent variable θ.

(iii) Give the equation that results in the case of an inverse square law $f(r) = -\mu r^{-2}$ and indicate briefly how this may be solved. [You are not required to solve the equation.] (15)

(b) An observer $\mathcal{O}'$ moves at a constant velocity $u \neq 0$ in the x-direction relative to another observer $\mathcal{O}$. In special relativity the coordinates of the event (x', t') observed by $\mathcal{O}'$ are related to the coordinates (x, t) of the same event observed by $\mathcal{O}$, by a Lorentz transformation:

$$x' = \gamma(x - ut), \quad t' = \gamma(t - ux/c^2), \quad \gamma(u) = \frac{1}{\sqrt{1 - u^2/c^2}}\,.$$

(i) Show that the space–time interval s^2 between the event $(0, 0)$ for both observers and the event at (x, t) for $\mathcal{O}$, (x', t') for $\mathcal{O}'$, is an invariant; in other words show that

$$s^2 = c^2t'^2 - x'^2 = c^2t^2 - x^2.$$

(ii) Using the result of part (i), show that two events at the same location $x' = 0$ separated by a time interval $t' > 0$ for $\mathcal{O}'$, are observed by $\mathcal{O}$ to be separated by a greater time interval $t > t'$.

(iii) Obtain an explicit formula for this effect of time dilation, by expressing t in terms of t' and γ. (10)

[**25**]